# Water Purification

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*Abstract*— Access to safe drinking water is a major global challenge due to pollution, climate change, and poor water management. Although technologies such as Reverse Osmosis, chlorination, and distillation are widely used, they suffer from high energy use, water wastage, mineral loss, and the lack of real-time monitoring. In Sri Lanka, these issues are worsened by groundwater contamination, high fluoride levels, CKDu prevalence, flood-related pollution, and aging infrastructure.

This research proposes a Smart Water Purification Ecosystem that integrates IoT, AI, and Machine Learning to provide an energy-efficient, intelligent, and adaptive water purification solution. The system enables real-time water quality monitoring, optimized purification, mineral balancing, and autonomous environmental assessment.

By combining real-time data, intelligent automation, and predictive maintenance, the proposed framework enhances reliability, reduces costs, minimizes water and energy waste, and supports sustainable water resource management for both urban and rural communities.[5]

## I. INTRODUCTION

Water is the most essential natural resource for human survival. However, rapid urbanization, agricultural expansion, industrial pollution, and climate change have significantly degraded water quality worldwide. Conventional water purification methods such as boiling, chlorination, Reverse Osmosis (RO), activated carbon filtration, and distillation are widely used, but each method has serious limitations including high energy usage, removal of beneficial minerals, chemical by-product formation, water wastage, and lack of automated quality verification.[6]

In Sri Lanka, a large portion of the population depends on groundwater, wells, and municipal water supplies, many of which are contaminated with fluoride, heavy metals, agrochemicals, and microbial pathogens. The Chronic Kidney Disease of Unknown Origin (CKDu) crisis in the North Central Province, post-flood contamination, and the increasing dependence on RO-based purification systems further emphasize the urgent need for an intelligent, sustainable, and energy-efficient water purification solution.[3]

To address these challenges, this research proposes a Smart Water Purification Ecosystem that integrates Internet of Things (IoT), Artificial Intelligence (AI), and Machine Learning (ML) technologies to deliver an end-to-end intelligent water purification, analysis, and enhancement platform. The proposed system is designed around the following four core research functions:

1. AI-Based Water Type Identification System

This function utilizes real-time sensor data such as pH, turbidity, total dissolved solids (TDS), and conductivity to automatically identify and classify different water types and determine the most suitable purification method.

2. Low-Power Household Distilled Water Generator with Mobile App Control

This function focuses on designing an energy-efficient household distillation system with IoT-based monitoring and mobile application control to ensure effective purification with minimal power consumption.[1][2]

3. Intelligent Alkaline Water Generation Using AI-Based Mineral Balancing

This function enhances purified or distilled water by precisely controlling mineral dosing using AI/ML techniques to maintain safe and healthy alkalinity levels suitable for human consumption.

4. Autonomous Water Testing Bot (WT Bot) for Real-Time Environmental Water Quality Testing

This function introduces a self-operating robotic system capable of collecting water samples from environmental sources such as rivers, wells, reservoirs, and flood-affected

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areas and transmitting real-time water quality data for early contamination detection and large-scale monitoring.
By integrating real-time monitoring, intelligent control, automated purification, mineral enhancement, and autonomous environmental surveillance into a unified ecosystem, the proposed solution aims to improve water safety, reduce energy and water wastage, enhance public health protection, and support sustainable water resource management. This research ultimately contributes toward the development of a next-generation smart water purification framework suitable for both urban and rural communities.

## II. LITERATURE REVIEW

### 1. Conventional Water Purification Technologies

Existing purification methods include RO, distillation, UV sterilization, chlorination, and carbon filtration. Studies show that RO systems remove most contaminants but also remove essential minerals and waste a significant portion of input water. Distillation produces extremely pure water but is highly energy-intensive and unsuitable for large-scale or daily household use. UV purification is effective only for microorganisms and does not remove heavy metals or chemicals. Chlorination produces harmful disinfection by-products such as trihalomethanes (THMs). Activated carbon filters improve taste but fail to remove pathogens and dissolved chemicals.[4]

### 2. Water Purification Challenges in Sri Lanka

Research conducted in Sri Lanka highlights groundwater contamination with fluoride, arsenic, cadmium, and agricultural pesticides, particularly in dry zone areas. The CKDu epidemic is suspected to be linked to long-term consumption of contaminated groundwater. Government-installed RO plants have helped reduce exposure but have introduced new problems such as high energy consumption, dependence on continuous maintenance, and mineral-deficient drinking water.

### 3. Alkaline Water and Health

Several studies suggest that alkaline water (pH 8–9.5) may help neutralize acid reflux, improve hydration, and supplement essential minerals such as calcium and magnesium. However, improper alkaline water preparation using chemical methods rather than mineral-based methods may lead to imbalanced water chemistry and potential health risks. Therefore, controlled and sensor-driven alkaline water generation is required.[6][7]

### 4. IoT in Water Quality Monitoring

IoT-based water monitoring systems are widely researched for real-time sensing of pH, turbidity, TDS, temperature, and conductivity. These systems enable remote monitoring, automated alerts, cloud data storage, and preventive maintenance. However, many existing solutions lack intelligent decision-making and predictive capabilities.

### 5. AI and Machine Learning in Water Systems

AI/ML is increasingly used in water quality prediction, contamination detection, filter life prediction, and consumption pattern analysis. ML classifiers can categorize water types based on sensor data and predict abnormal conditions before they become hazardous. However, limited research integrates AI-based water classification with automated mineral balancing for alkaline water.

### 6. Autonomous Water Sampling Systems

Robotic and drone-based water sampling systems are emerging for environmental monitoring of rivers, lakes, and reservoirs. These systems provide high-resolution spatial data for pollution mapping and early contamination detection. However, integration of such systems with household purification units remains limited.

**Research Gap Identified**

From the literature, it is evident that:

- Current systems lack end-to-end intelligence.
- No single system integrates AI-based water identification, energy-efficient distillation, intelligent alkaline mineral balance, and autonomous environmental water testing.
- There is a significant gap in adaptive purification systems designed for Sri Lankan environmental conditions.
- This research addresses these limitations.

## III. REFERENCES


[1] Google – Distilled Water
[2] Nicole Blades, Medically Reviewed by Kathleen M. Zelman, RD, LD, MPH, “Distilled Water” September 15, 2024
[3] Medically Reviewed by Shruthi N, MD, “Water Quiz: How Much Do You Know About Hydration”, October 23, 2024
[4] Jamal Qureshi, “Svalbardi Polar Iceberg Water”
[5] Google - What is distilled water used for at home?
[6] Google - distilled water vs ph7 water vs alkaline water
[7] Medically reviewed by Angelica Batingit, MD – Written by Rena Goldman, “Overview of Alkaline water, and Where to Gelt it”, Updated on November 14, 2024